Controlled Evaluation of Class-Imbalance Mitigation Across Histopathology Patch Classification and WSI-Level MIL

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Abstract

This study evaluates methods for mitigating class imbalance in computational pathology, covering patch classification and whole-slide multiple-instance learning (MIL) in CAMELYON16, TCGA-UT, BRACS, and PANDA, all represented with frozen Virchow2 features. Patient-disjoint splits are created before imbalance is manipulated, and only training support changes: a full natural anchor is accompanied by size-matched balanced, moderate ($\rho = 10$), and severe ($\rho = 100$ where feasible) conditions. Validation and test patients, held-out prevalence, training-set size, model family, and numbers of optimizer updates remain fixed. Patch conditions retain a common patient and slide pool while changing patch allocation; MIL conditions vary slide support and report the resulting change in patient diversity. The primary discrimination endpoint is balanced accuracy, evaluated alongside a co-primary tail-calibration endpoint. Mitigation recovery is assessed only when prespecified thresholds show a meaningful imbalance-induced deficit on the corresponding endpoint. Secondary analyses address classwise discrimination, tail-sensitive calibration, computational cost, intrinsic separability, and condition-specific learnability.

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1 Introduction

Class imbalance changes the empirical objective faced by a classifier: frequent classes supply more optimization terms, while rare classes may contribute too little evidence to establish stable decision boundaries. In pathology, the problem occurs at two distinct prediction units. Patch classification uses directly labelled crops, whereas WSI-level MIL predicts from bags whose instances inherit only a weak slide label. The same slide cohort can therefore be nearly balanced between diagnoses while containing very sparse class-informative tissue. Aggregate accuracy does not isolate either failure mode, motivating class-balanced discrimination and probability-quality evaluation (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Native-dataset comparisons do not identify an imbalance effect by themselves. They simultaneously vary task, class separability, training support, label granularity, and evaluation prevalence. This study instead changes class allocation only in the training partition and retains identical natural validation and test cohorts. A size-matched balanced CE condition estimates attainable performance at the same training budget, so mitigation can be interpreted as recovery of damage attributable to the imposed imbalance rather than as an unconstrained gain over a convenient baseline.

The study addresses three research questions:

RQ1. Effect of imbalance. At fixed task, patient split, encoder, training-set size, and evaluation set, how does increasing training imbalance affect tail-class discrimination and calibration in patch classification and WSI-level MIL?

RQ2. Mitigation effectiveness. Which mitigation methods recover performance lost specifically to training imbalance, and what are their calibration and computational costs?

RQ3. Predictors of damage and mitigation headroom. Across all controlled comparisons, do intrinsic class separability, condition-specific learnability, and independent support predict whether and how strongly imbalance damages CE performance? Conditional on demonstrated damage, do the same quantities predict mitigation recovery?

RQ1 establishes whether a recoverable problem exists, and RQ2 asks which methods address that demonstrated deficit. RQ3 first models CE damage across every controlled comparison and then models recovery only where the corresponding deficit is large and precise enough to interpret. The datasets and preprocessing provenance are documented in the companion dataset report, and all detailed method objectives and variants are defined in the companion methods report (Queisler, 2026a,b).

2 Experimental Design

A controlled imbalance experiment is informative only when the balanced and imbalanced models differ in their training class allocation rather than in their data source, representation, or evaluation cohort. The design therefore establishes the patient split, frozen features, and held-out examples once, then constructs several training distributions at a common support budget. The following subsections define the two prediction regimes, explain how their evidence is controlled, and specify how mitigation methods are trained and compared.

2.1 Datasets and prediction regimes

Table 1 defines the prediction example and label in each available dataset–regime. The terms *patch* and *tile* both denote a 256×256 image crop; *patch* is used throughout this report. A patch-level example is one such crop with the label specified in the second column. A bag-level example is one WSI represented by frozen embeddings of its eligible patches and receives one label for the entire WSI.

Every eligible patch embedding enters the MIL model for its WSI, so bag size reflects the tissue that remains after the dataset-specific eligibility pipeline. The complete bag for a given slide is identical across training conditions and methods.

Before examples are selected, each patient or case is assigned to exactly one *partition*: the training, validation, or test cohort. All slides from the same patient follow that assignment. Dataset–regime eligibility is also fixed before splitting. A sample is excluded if it lacks a valid target or split-unit identifier, yields no patch under the dataset-specific tissue rule, lacks an annotation required to assign a patch label, or lacks readable frozen features. The same rules are applied before constructing every imbalance condition.

Dataset	Patch-level example and label	MIL bag and bag label	Split unit
CAMELYON16	Non-overlapping tissue crop at 20x; tumor if at least 50% of its aligned mask cell is tumor, normal otherwise	One WSI bag; metastasis-bearing (tumor) versus normal slide	Slide (one slide per patient)
TCGA-UT	Author-supplied tumor-region crop; inherits its source WSI's cancer type	One diagnostic-WSI bag; the same one of 32 cancer types	TCGA participant
BRACS	Non-overlapping crop from a pathologist-annotated ROI; inherits one of seven ROI subtypes	One WSI bag; official seven-class WSI label defined by the most aggressive lesion detected in the slide	Patient
PANDA	Foreground crop; cancer if at least 50% of its aligned provider-specific mask cell is cancer, benign otherwise	One biopsy-WSI bag; ordinal ISUP grade 0–5	Biopsy (one WSI)

Table 1: Dataset-specific definitions of patch-level examples and bag-level examples. Only TCGA-UT uses the same target in both regimes. CAMELYON16 and PANDA change diagnostic granularity, while BRACS changes from the annotated ROI subtype to the official most-aggressive-lesion WSI subtype.

Patch classification. Eligible patches are embedded once with frozen Virchow2. Concatenating the CLS token and mean patch token gives 2560 input features (Vorontsov et al., 2024; Zimmermann et al., 2024). For CAMELYON16 and PANDA, a patch label is derived from the spatially aligned expert mask and can differ from the WSI label. A TCGA-UT patch inherits the cancer-type label of the diagnostic WSI from which the dataset authors cropped it. A BRACS patch inherits the subtype of its pathologist-annotated ROI. All patches that pass these dataset-specific eligibility and label rules enter the eligible pool before condition-specific training subsampling. The common classifier is a two-layer multilayer perceptron (MLP) with a 512-unit hidden representation, rectified linear unit (ReLU) activation, dropout, and a linear class output. CFAL alone replaces the linear output with a prototype-based affinity head; OKO adds a training-only auxiliary head that is discarded for evaluation.

WSI-level MIL. A slide is one bag of frozen patch embeddings; here, an *instance* means one patch representation inside that bag, not a separately labelled training example. CAMELYON16 bags use the slide diagnosis, TCGA-UT bags use the diagnostic WSI's cancer type, and PANDA bags use the biopsy's ordinal ISUP grade. A BRACS bag uses the official seven-class WSI annotation, which the dataset defines from the most aggressive lesion detected in the slide. This label is read from the official WSI metadata rather than reconstructed from ROI annotations, so WSIs without released ROIs remain eligible and no project-specific severity ordering can alter the target. Eligible tissue instances are selected by the same deterministic pipeline used for the other WSI datasets. The common attention-MIL model maps all instances to a

256-dimensional space, assigns normalized attention weights, and classifies their weighted average. SC-MIL adds a training projection head for its contrastive objective, and MDE-inspired MIL replaces the single classifier with two expert heads. Slide labels are the only supervision; patch labels, ROI labels, and masks are not used by the primary MIL training runs (Brancati et al., 2022; Queisler, 2026a).

Comparative scope. Only TCGA-UT permits a direct patch-versus-WSI comparison. Both regimes predict the same cancer-type label within the same participant-disjoint cohort. In CAMELYON16, patch classification detects tumor tissue, whereas a WSI bag is classified as metastasis-bearing or normal. In PANDA, patch classification detects cancer tissue, whereas a WSI bag predicts the biopsy's ISUP grade. In BRACS, a patch predicts its ROI-specific subtype, whereas a WSI bag predicts the official most-aggressive-lesion subtype for the complete slide. The CAMELYON16, PANDA, and BRACS regimes are therefore separate dataset–target groups rather than paired measurements of one task.

2.2 Controlled comparison

The construction asks a simple counterfactual question: how would the same prediction task perform if a fixed training budget were distributed evenly across classes rather than concentrated in a head class? Answering this question requires more than specifying an imbalance ratio. The comparison must first fix the patient split, eligible examples, representation, evaluation cohort, and total controlled training support; only then can class allocation be changed. Figure 1 summarizes this separation between fixed design choices and the intended intervention.

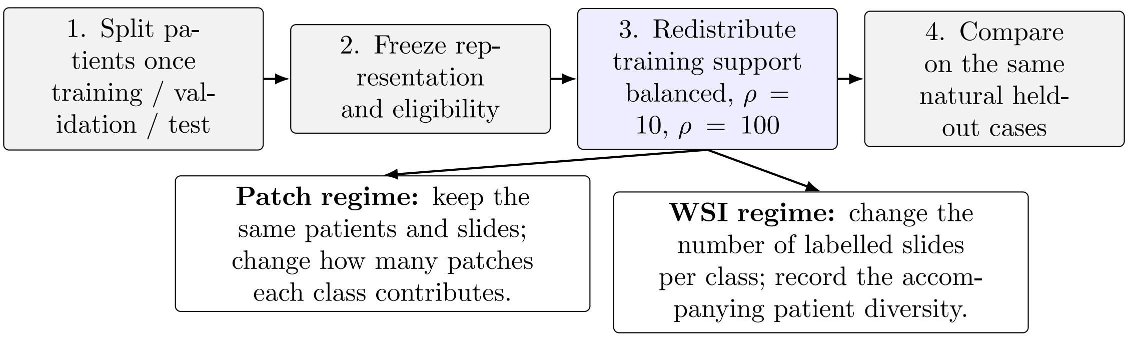

Figure 1: Controlled construction of the benchmark. Gray steps are fixed within a split; the blue step is the intended intervention. Patch imbalance redistributes patches within a common patient and slide pool, whereas WSI imbalance changes labelled slide support.

Each eligible dataset–regime–split has four training conditions:

1. the full natural training set, used only as a descriptive anchor;
2. a size-matched balanced training set;
3. a moderate imbalance condition targeting $\rho = 10$; and
4. a severe imbalance condition targeting $\rho = 100$ where feasible.

The balanced, moderate, and severe conditions contain the same total training support T: patches for patch classification and slides for MIL. The largest feasible shared T is chosen from training data alone under the support constraints defined below. The natural anchor may contain a different total and is excluded from the imbalance deficit and recovery estimands in Section 3.2. Thus contrasts among the three controlled conditions change class allocation without also changing the number of training examples.

The target ratio $\rho_{\text{target}} \in \{1, 10, 100\}$ specifies the intended intervention, but integer allocation and finite inventories can change the realized class distribution. The achieved ratio is therefore recomputed from every manifest. For realized class supports $N_1, \ldots, N_K$, it is

$$\rho_{\text{achieved}} = \frac{N_{\max}}{N_{\min}}. \tag{1}$$

This quantity records the extreme head–tail contrast. The complementary normalized-entropy deficit

$$1 - H_{\text{norm}} = 1 - \frac{-\sum_{c=1}^{K} p_c \log p_c}{\log K}, \qquad p_c = \frac{N_c}{\sum_j N_j}. \tag{2}$$

uses all classes and is normalized for class count: it is zero for a balanced allocation and increases as support becomes concentrated. Both quantities are computed separately at patch and slide level from each realized training manifest. Per-class patient, slide, and patch counts are reported alongside them, because neither scalar reveals whether many patches originate from only a few patients.

Patient-disjoint training, natural-validation, and natural-test partitions are created first. Imbalance manipulation is confined to training support. Within a split, every condition uses exactly the same validation patients, test patients, and eligible held-out examples. No held-out example is deleted, duplicated, or reweighted according to the training severity. Patch-level evaluation uses every eligible held-out patch, and slide-level evaluation uses every eligible held-out slide. Consequently, recall is estimated from all available held-out cases and prevalence-dependent precision and macro F1 are compared at one fixed test prevalence.

The validation cohort also retains its natural distribution. It is used for hyperparameter and checkpoint selection but never resized to resemble a training condition. Training, validation, and test manifests are versioned with patient identifiers, class counts, construction seed, and content hashes before model fitting.

2.3 Independent evidence and support floors

A patch is a training example, but it is not independent evidence when thousands of patches come from the same patient or slide. In this report, *independent support* therefore means the number of distinct patients and slides that underlie a class's training examples. Controlling this support prevents an apparent patch-level head–tail contrast from being merely a contrast between many correlated crops from one patient and fewer crops from another.

The rule is adapted to the prediction regime. For patch classification, a fixed per-class patient pool and, within it, a fixed slide pool are shared by the balanced and all imbalanced conditions. Sampling visits patients, then slides within patients, and only then patches. No patient may supply more than 10% of a class's patch allocation and no slide more than 5%; deterministic round-robin selection enforces these caps before any unit is revisited. Patch imbalance can therefore change the number of patches per class without changing which patient and slide pools are available. For MIL, a slide is itself the labelled example, so reducing slide support necessarily reduces independent support. Selection visits patients before taking additional slides from the same patient; each slide contributes once, and no patient supplies more than 10% of a class's slides. MIL contrasts are consequently interpreted as controlled slide-support imbalance, including its induced change in patient diversity. Every manifest reports per-class patient, slide, and patch counts, maximum contributions, and the fraction of the eligible independent pool retained.

The smallest defensible tail support is estimated before the definitive comparison with a *support pilot*. These balanced CE runs ask when additional independent training units cease to materially improve natural-validation performance; they do not tune a hyperparameter or contribute observations to the definitive analysis. In the patch regime, one pilot unit is a patient, or a slide where patient and slide coincide. Each selected unit contributes the same fixed quota q_{patch} of patches, distributed as evenly as possible over its eligible slides. The quota is determined from the training inventories at the largest candidate level and then held fixed, so increasing a pilot level adds patients rather than patches per patient. In the MIL regime, one pilot unit is a labelled slide represented by all its eligible instances.

Nested pilots compare $5, 10, 15, 20$, and 30 units per class, or all available units when the scarcest class has fewer. They are repeated for three independently seeded *construction orderings*. An ordering is a fixed sequence of eligible units; its prefix of length m is simply the first m units in that sequence. Thus the 10-unit pilot retains the same first five units as the five-unit pilot and adds the next five. At lower levels of the hierarchy, every patient has a fixed slide order and every slide has a fixed patch order. Figure 2 illustrates this nesting.

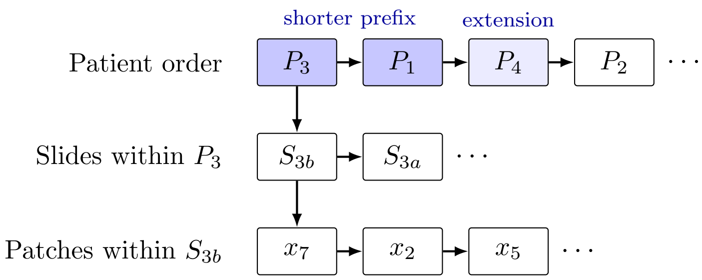

Figure 2: Hierarchy behind one construction ordering. A seed fixes the patient sequence and the local slide and patch sequences. The darker patient boxes form a shorter prefix; the lighter third box is added, without replacing earlier units, when the prefix is extended.

Because every larger pilot level extends the preceding prefix, adjacent levels are paired within an ordering and evaluated on the same natural-validation cohort. The smallest level whose next increment changes mean balanced accuracy by less than 0.01 and every class recall by less than 0.02 in all three orderings is the *stability floor*; if none qualifies, the largest candidate is used.[1]

The definitive minimum is the larger of this empirical stability floor and a prespecified design floor. Each class must contain at least 10 patients and 20 slides, or 20 slides when patient and slide coincide, and any class-aware batch or set must contain at least two distinct same-class units. The 10-patient and 20-slide values are not estimated performance thresholds: they are the smallest counts compatible with the 10% patient and 5% slide contribution caps, respectively. The same 20-slide lower bound is retained for MIL to keep slide-based method comparisons outside a five-example regime even though MIL has no separate 5% slide cap. A requested severity is reduced to the largest allocation satisfying both floors; a dataset–regime is excluded if even its balanced condition fails them. Pilot quotas, curves, seeds, and resulting floors are frozen and reported with the definitive manifests.

2.4 Severity profiles and tail assignments

A binary task needs only a head and a tail count, but a multiclass task also needs a rule for the classes between those extremes. The exponential construction places the K class supports at equal intervals on a logarithmic scale. For a locked order from head ($i = 0$) to tail ($i = K - 1$), it assigns weights

$$w_i(\rho_{\text{target}}) = \rho_{\text{target}}^{-i/(K-1)}, \qquad n_i^* = T \frac{w_i(\rho_{\text{target}})}{\sum_{j=0}^{K-1} w_j(\rho_{\text{target}})}, \quad i = 0, \ldots, K-1. \tag{3}$$

[1]The five-unit pilot is the sole cap exception. With only five contributing patients, at least one patient must supply at least 20% of a class, so a 10% cap is mathematically impossible; a 5% slide cap also requires at least 20 contributing slides. The patch pilot therefore uses five patients with equal quotas, and the MIL pilot uses five slides from five patients. This diagnostic pilot level is never a definitive training allocation.

so the head-to-tail ratio is exactly ρ_{target} before integer rounding. For example, with four classes and $\rho_{\text{target}} = 100$, the unnormalized weights are approximately $1, 0.215, 0.046$, and 0.01; the two body classes lie geometrically between the head and tail instead of being assigned ad hoc counts. Binary targets use $n^*_{\text{head}}/n^*_{\text{tail}} = \rho_{\text{target}}$ directly. A prespecified step profile may be used for clinically meaningful BRACS groups, with its grouping and realized supports reported before analysis.

The values n^*_i are real-valued targets, whereas a manifest must contain whole patches or slides. *Constrained integer allocation* converts them into usable class counts. Each target is rounded to the nearest integer and clipped to the class's support floor and available unique support. If the rounded counts do not sum to T, units are added to or removed from the classes with the largest rounding residuals until the total is exact. The resulting integers n_i therefore satisfy $\sum_i n_i = T$ and every class-specific lower and upper bound while remaining close to the exponential targets under this deterministic adjustment.

Counts determine how many units a class contributes; construction orderings determine which units they are. One master ordering is created per class at each hierarchy level. Every condition selects a deterministic prefix, or a patient-round-robin extension of a prefix, from these same orderings. For patch classification, all fixed-pool patients and slides contribute before additional patches are allocated; for MIL, slides are selected by cycling through the patient prefix before revisiting a patient.

If a target ratio is infeasible, the same rule is applied to every dataset and regime: use the largest attainable ratio and report the requested ratio, achieved ratio, limiting class, realized counts, and binding support floor. Validation or test support is never reduced to manufacture feasibility. This general fallback is especially relevant to high-K tasks, where an exponential profile must allocate support to many intermediate classes as well as the tail; it is not a dataset-specific exception. A stable lower-severity condition is preferred to a nominal $\rho = 100$ condition supported by only a few independent units.

The profile determines support by rank, so a second decision maps semantic classes to those ranks. Using only one mapping would confound a class's intrinsic difficulty with its position in the tail. Multiclass targets therefore use three locked assignments where feasible: (i) classes ordered by clinical meaning or native support, (ii) the reverse of that order for ordinal targets or a one-position rotation for nominal targets, and (iii) one distinct random permutation drawn and recorded before fitting. Figure 3 illustrates the nominal case. Binary tasks use both head–tail orientations. If support constraints make an assignment infeasible at the shared T, the achieved allocation is reported and the assignment is omitted only under a prespecified feasibility rule.

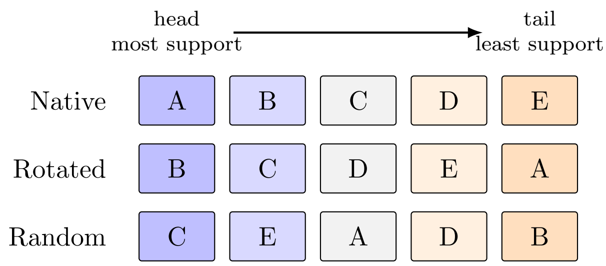

Figure 3: Illustrative tail assignments for a five-class nominal target. Columns are support ranks fixed by the severity profile; rows change which semantic class occupies each rank. Ordinal targets reverse rather than rotate the second assignment.

The random quantities are separated into five seed families so that changing one source of variation does not silently change another. Table 2 defines their roles.

Seed family	Determines	Prespecified use
Patient split	Membership of the training, validation, and test cohorts	Three locked patient-disjoint splits
Tail assignment	Mapping of semantic classes to head–tail support ranks	Recorded clinical/native, reversed/rotated, and random mappings
Construction	Patient, slide, and patch orderings and the selected evidence	Three pilot orderings and a disjoint fourth ordering for definitive manifests
Initialization	Parameter initialization and optimization randomness	Two tuning seeds and five disjoint confirmation seeds
Resampling	Bootstrap draws and permutation swaps	Uncertainty intervals and hypothesis tests only

Table 2: Seed families and their non-overlapping roles in the benchmark.

No seed is chosen based on test performance. The size-matched balanced manifest and its trained CE baseline are shared by all tail assignments within a split.[2] The same balanced predictions enter every assignment-specific deficit and are represented once in resampling data, avoiding duplicated balanced fits, bootstrap records, and artificial assignment variability.

2.5 Mitigation methods

Table 3 lists only the methods evaluated in this benchmark. CE and MIL CE are the size-matched no-mitigation references. Method objectives, tested adaptations, ablations, defaults, and fidelity qualifications are defined exclusively in the companion methods report (Queisler, 2026b).

Family	Tested methods	Regime
No-mitigation reference	CE; MIL CE	Patch; WSI
Data-level sampling and augmentation	Balanced sampling; RankMix-inspired feature mixing	Both for sampling; WSI for RankMix-inspired
Losses and prior correction	Weighted CE; focal loss; train-time and post-hoc logit adjustment; CFAL	Both, except CFAL patch only
Hybrid sampling–loss	CE + soft F1; CE + soft MCC; OKO set learning; MDE-inspired dual-expert MIL	Patch for CE hybrids and OKO; WSI for MDE-inspired
Representation and architecture	cRT; SC-MIL	Both for cRT; WSI for SC-MIL

Table 3: Prespecified method roster, using the same intervention-level taxonomy as the companion methods report. "Both" means the patch and WSI implementations defined there, not a comparison of their raw scores.

2.6 Training, selection, and replication

The frozen encoder, classifier family, optimizer family, batch definition, augmentation, and numerical precision are held fixed within a regime. Training is budgeted only in optimizer updates; epochs are neither a stopping rule nor a unit of comparison. Let B be the regime's locked batch size and T the common controlled support. A single-stage model receives

$$U = 30 \left\lceil \frac{T}{B} \right\rceil \tag{4}$$

updates, corresponding to 30 reference passes through T examples under ordinary sampling. Balanced sampling, replacement, synthetic bags, and dual loaders do not change U. RankMix

[2] At $\rho_{\mathrm{target}} = 1$, every class receives equal support. Permuting semantic classes across support ranks therefore leaves the allocation unchanged, so tail assignment is not a meaningful factor for the balanced condition.

receives U teacher updates and, after reinitialization, U student updates. cRT receives U stage-one updates and $\lceil 0.2U \rceil$ stage-two classifier updates. MDE performs U joint updates, each consuming its prespecified natural and balanced minibatches. The natural anchor uses the same formula with its own support and is descriptive. Processed examples, unique-example exposures, and updates are all reported. Each run stores validation metrics at fixed update intervals. The selected checkpoint maximizes natural-validation balanced accuracy, with macro F1 and then lower NLL as deterministic tie-breakers. Test predictions are produced once from that locked checkpoint.

Hyperparameter comparisons are bounded by a common validation-search budget rather than by restricting every method to one tuned quantity. For each trainable mitigation method, its four-value imbalance-specific grid is crossed with the common four-value learning-rate grid in Appendix A, yielding at most 16 candidate configurations. This joint search accommodates interactions between a method's objective and optimization dynamics while holding the optimizer family, weight decay, architecture, batch definition, and all other training parameters fixed. CE and methods without a meaningful imbalance-specific control evaluate the four learning rates without introducing an artificial search dimension. Train-time logit adjustment is tuned jointly over τ and learning rate; post-hoc logit adjustment inherits the selected CE model and tunes only τ because it performs no gradient optimization.

Every hyperparameter candidate is fitted with the same two tuning initialization seeds for each prespecified patient split and tail assignment. Selection is performed separately for each method, dataset, prediction regime, and imbalance severity. Within each such selection unit, natural-validation balanced accuracy is averaged across both tuning seeds, all tuning splits, and all tail assignments. The configuration with the highest aggregate value is selected; macro F1 and then lower NLL are deterministic tie-breakers. The chosen configuration is locked before any of the five disjoint confirmation initialization seeds are fitted and before test evaluation. Tuning runs never enter confirmation effect estimates. For cRT, stage one inherits the selected CE configuration, while the stage-two classifier learning rate is selected separately from the common grid. Temperature scaling, when reported, fits one positive scalar on natural-validation NLL after model and checkpoint selection and applies it unchanged to test logits (Guo et al., 2017).

A *controlled contrast* is the difference between two matched runs, such as balanced CE versus imbalanced CE or a mitigation method versus imbalanced CE. The two runs use the same patient split, tail assignment, confirmation-seed index, and held-out cases, so the contrast does not mix the intended change with a different cohort or replication. Each locked patient split and tail assignment is repeated with the five confirmation seeds. Exactly three patient-split seeds are used; a dataset–regime unable to produce all three valid splits under the independent-support floors is excluded from the corresponding confirmatory analysis. The split counts, support floors, condition manifests, tuning and confirmation seeds, method grids, and update budgets are frozen in the signed analysis manifest before the first definitive fit. The unit of replication and all missing or failed runs are reported. A failed run is not silently replaced using a test-informed seed.

Computational cost is separated into validation search and locked-configuration fitting. Resource use is summarized by wall-clock training time, accelerator-hours, and peak accelerator memory. Model size is reported as total and trainable parameter counts, while data exposure is reported as effective passes through unique training examples. Costs include every method-specific stage, including both cRT stages, but exclude the one-time frozen-feature extraction shared by all methods. Hardware and software environments are recorded with each run.

2.7 Scope and limitations

The design isolates training-distribution effects under frozen Virchow2 representations; it does not establish how end-to-end encoder fine-tuning would respond to imbalance. Constructed tails improve causal interpretability but need not reproduce referral pathways or biological prevalence. A shared T controls data volume at the cost of discarding some available training examples, which is why the full natural condition is retained as a descriptive anchor. Some methods change more than one mechanism, so the study supports method-level comparisons rather than universal causal claims about sampling or loss design. The locked update budget equalizes optimizer steps but not information exposure: RankMix consumes teacher and student passes, MDE draws a natural and a balanced minibatch per update, and post-hoc logit adjustment performs no gradient optimization at all, so method comparisons are made at equal update budget with exposure and cost reported alongside, not at equal data throughput. MIL instance counts likewise vary with the eligible tissue area of each selected slide; processed instances, memory, and runtime are therefore reported rather than equalized by truncating bags. The number of independent patients, not the number of patches, limits inference. All imbalance-robustness findings are additionally conditional on the geometry of a single frozen encoder; whether the same deficits and recoveries hold under a different foundation representation is outside the present scope. Finally, calibration on a natural held-out cohort is conditional on that cohort's prevalence and should not be transported to a new deployment population without explicit prior-shift analysis.

3 Evaluation and Statistical Analysis

Evaluation separates predictive discrimination, probability quality, mitigation recovery, and exploratory predictors of damage and recovery. All estimands and inferential families are defined before test evaluation.

3.1 Endpoints and aggregation units

The primary endpoint is balanced accuracy, equal to macro recall for single-label multiclass prediction. Secondary discrimination outcomes are macro F1; per-class recall and F1; and head, body, and tail recall under the locked assignment. Classes are ranked by allocated training support: the first $\lceil K/3 \rceil$ ranks are head, the last $\lceil K/3 \rceil$ are tail, and intervening ranks are body; binary tasks have one head and one tail and no body. Ties follow the locked class ordering. Overall accuracy is descriptive. Patch predictions are summarized in two ways: patch-micro estimates treat eligible patches as predictions while uncertainty remains patient-clustered, and slide/patient-macro estimates first aggregate the metric contribution within each slide or patient so that heavily tiled slides do not dominate. WSI outcomes are slide-level with patient clustering where patients contribute multiple slides.

Let N denote the number of held-out predictions in an aggregation unit, C the number of classes, $y_i \in \{1, \dots, C\}$ the true label, and $\hat{\mathbf{p}}_i \in [0, 1]^C$ the predicted probability vector. Write Y_i for the one-hot encoding of y_i. Probability quality is assessed at natural-prevalence and class-balanced weightings.

Ordinal agreement and error. PANDA WSI labels are ordered ISUP grades, so an error of one grade should count less than an error spanning several grades. Let O_{ab} be the observed confusion-matrix count for true grade a and predicted grade b. The count expected under independent true and predicted grades is $E_{ab} = O_{a\cdot}O_{\cdot b}/N$, which retains the observed marginal grade frequencies. With $a, b \in \{0, \dots, C-1\}$ and quadratic disagreement weight $w_{ab} = (a - b)^2/(C-1)^2$, quadratic-weighted kappa is

$$\kappa_{\mathrm{QW}} = 1 - \frac{\sum_{a=0}^{C-1} \sum_{b=0}^{C-1} w_{ab} O_{ab}}{\sum_{a=0}^{C-1} \sum_{b=0}^{C-1} w_{ab} E_{ab}}. \tag{5}$$

A value of 1 denotes perfect grade agreement, 0 is the agreement expected from the marginal grade frequencies, and negative values indicate more weighted disagreement than expected by chance. Ordinal mean absolute error (MAE) is

$$\mathrm{MAE}_{\mathrm{ord}} = \frac{1}{N}\sum_{i=1}^{N} |g_i - \widehat{g}_i|, \tag{6}$$

where g_i and $\widehat{g}_i$ are the true and predicted ISUP grades. It reports the average distance in grade levels and is lower-is-better. Both measures are secondary outcomes for PANDA; kappa emphasizes agreement beyond the observed marginals, whereas MAE has the direct interpretation of grades missed per biopsy.

Negative log likelihood. Natural-prevalence negative log likelihood (NLL), also called multiclass log loss, is

$$\mathrm{NLL} = -\frac{1}{N}\sum_{i=1}^{N} \log \hat{p}_{i,y_i}. \tag{7}$$

Lower NLL indicates that more probability mass is assigned to the true class. The implementation clips $\hat{p}_{i,y_i}$ to $[10^{-12}, 1]$ before taking the logarithm. To prevent common classes from dominating probability-quality summaries, class-conditional NLL and macro NLL are

$$\mathrm{NLL}_c = -\frac{1}{N_c}\sum_{i:y_i=c} \log \hat{p}_{ic}, \qquad \mathrm{NLL}_{\mathrm{macro}} = \frac{1}{C}\sum_{c=1}^{C} \mathrm{NLL}_c, \tag{8}$$

where N_c is the number of held-out predictions from class c. Natural-prevalence NLL is the deployment-facing summary for the observed held-out cohort; macro NLL and unweighted head, body, and tail averages of NLL_c are tail-sensitive summaries.

Multiclass Brier score. The multiclass Brier score penalizes squared error across the full predicted distribution (Brier, 1950):

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{p}}_i - Y_i\|_2^2\,. \tag{9}$$

Lower values indicate better probability quality. The natural-prevalence score averages this error over the complete held-out cohort. Classwise scores apply the same full-distribution error only to predictions with $y_i = c$. Their unweighted averages provide head, body, and tail summaries.

Expected calibration error. For each prediction, let $\hat{y}_i = \arg\max_c \hat{p}_{ic}$ and $q_i = \max_c \hat{p}_{ic}$. Expected calibration error (ECE) partitions confidence into $B = 10$ fixed equal-width bins on $[0, 1]$ and compares mean confidence with empirical accuracy in each nonempty bin (Naeini et al., 2015):

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{N} \left| \frac{1}{|I_b|}\sum_{i\in I_b} \mathbb{1}[\hat{y}_i = y_i] - \frac{1}{|I_b|}\sum_{i\in I_b} q_i \right|, \tag{10}$$

where I_b contains the predictions assigned to bin b. ECE is secondary because it depends on binning and can be unstable with small classwise samples; its bootstrap uncertainty uses the same fixed bins and patient weights as the other endpoints. Reliability diagrams show the binwise confidence–accuracy relation overall and for the tail group.

Because checkpoints and hyperparameters are selected for natural-validation balanced accuracy, all probability-quality results describe discrimination-selected models rather than models optimized for calibration. Temperature scaling and target-prior correction are the only calibration-specific adjustments applied after selection.

Scores for balanced decisions are kept distinct from probabilities for the natural target prevalence. For an ordinary CE score z_c^{CE} and realized training prior π_c^{train}, post-hoc logit adjustment uses $z_c^{\mathrm{CE}} - \tau \log \pi_c^{\mathrm{train}}$ for balanced-accuracy decisions. It is not reported as a naturally calibrated probability. Under the prior-shift assumption, target-prevalence probabilities instead normalize

$$z_c^{\mathrm{CE}} - \log \pi_c^{\mathrm{train}} + \log \pi_c^{\mathrm{target}}. \tag{11}$$

For train-time logit adjustment, training applies CE to $g_c = z_c + \tau \log \pi_c^{\mathrm{train}}$. Because g_c approximates training-posterior logits, the corresponding target-prevalence logits are

$$z_c + (\tau - 1) \log \pi_c^{\mathrm{train}} + \log \pi_c^{\mathrm{target}}. \tag{12}$$

The unadjusted z_c remains the train-time method's validation-selected discrimination score but is a fully prior-stripped score only when $\tau = 1$. The target prior π^{target} is estimated separately within each natural-validation split and applied only to the corresponding test split. Temperature scaling is fitted on natural-validation NLL after the appropriate target-prior correction and applied unchanged to test logits. Raw, balanced-decision, target-prior-corrected, and temperature-scaled outputs are retained separately. A single temperature is interpreted as a sharpness correction, not a remedy for class-specific prior mismatch.

3.2 Imbalance deficit, recovery, and inference

Recovery is meaningful only when imbalance has first caused a measurable loss. The analysis therefore proceeds in two stages: compare size-matched balanced and imbalanced CE, then evaluate mitigation only on an endpoint that shows sufficiently large and statistically supported harm. One *comparison unit* is a fixed dataset, prediction regime, imbalance severity, and tail assignment. A *gate* is the prespecified eligibility rule that routes such a unit to the relevant recovery analysis; it is not a learned model component.

For a higher-is-better metric M, define the imbalance-induced deficit and recovery fraction at each imbalanced condition as

$$\begin{aligned} D_M &= M(\text{balanced CE}) - M(\text{imbalanced CE}), \\ R_M &= \frac{M(\text{method on imbalanced data}) - M(\text{imbalanced CE})}{D_M}. \end{aligned} \tag{13}$$

$R_M = 0$ denotes no recovery, $R_M = 1$ recovery to the size-matched balanced CE reference, $R_M < 0$ further deterioration, and $R_M > 1$ performance beyond that reference. Because the balanced reference draws more unique tail independent units than any imbalanced condition at the same T, methods that only reweight or recalibrate the imbalanced manifest cannot restore tail evidence that was never sampled; for that family $R_M < 1$ may reflect an information ceiling rather than method failure, whereas resampling and synthesis methods are the only family for which $R_M = 1$ is mechanically attainable. The full natural condition is descriptive and never enters D_M or R_M.

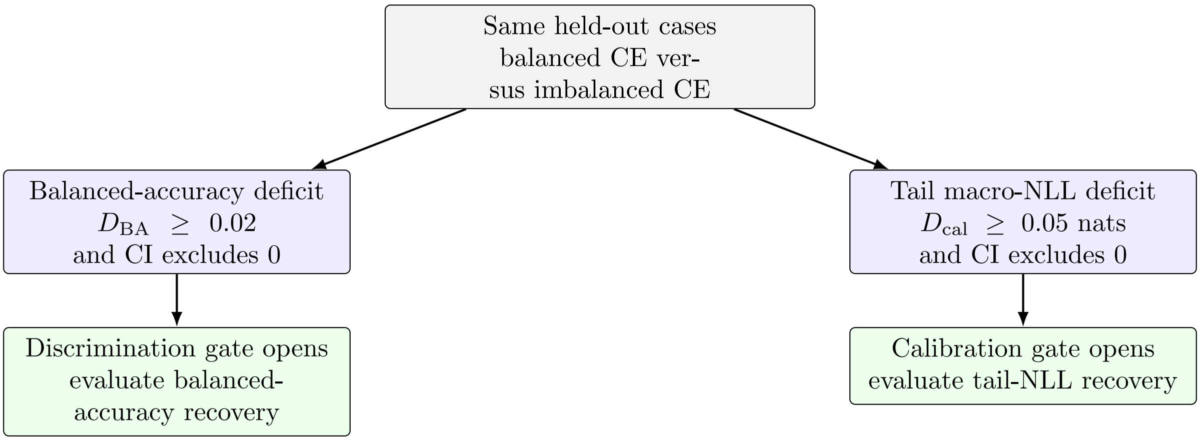

Figure 4: Routing from an imbalance deficit to mitigation analysis. A comparison unit can enter through either gate or both. If neither rule is met, its CE deficits are still reported, but a recovery ratio is not interpreted. CI denotes the paired 95% confidence interval.

The *discrimination gate* opens when the CE balanced-accuracy deficit is at least 0.02 and its paired 95% confidence interval excludes zero. The *calibration gate* opens when the CE tail-group macro-NLL deficit, $\mathrm{NLL}_{\mathrm{tail}}(\text{imbalanced CE}) - \mathrm{NLL}_{\mathrm{tail}}(\text{balanced CE})$, is at least 0.05 nats with a paired 95% confidence interval excluding zero. Both rules are computed from CE runs on the same held-out cases before any mitigation method is compared. A comparison unit entering through the discrimination gate is evaluated for balanced-accuracy recovery; one entering through the calibration gate is evaluated for tail-calibration recovery; it may enter through both.

The two routes preserve an important distinction. Strong frozen features may leave balanced accuracy nearly intact while imbalance still biases tail probabilities, in which case a single discrimination rule would exclude methods designed to correct class priors. Conversely, the gates avoid unstable recovery ratios when no meaningful damage exists to recover. All CE deficits on both axes remain reported, including units that fail both rules. For the lower-is-better calibration axis, the sign-corrected deficit $D_{\mathrm{cal}} = \mathrm{NLL}_{\mathrm{tail}}(\text{imbalanced CE}) - \mathrm{NLL}_{\mathrm{tail}}(\text{balanced CE})$ is the denominator and the method's tail-NLL improvement over imbalanced CE is the numerator, so Equation (13) retains the same interpretation. Recovery for other secondary metrics is descriptive and does not reopen a comparison unit that fails both gates.

Confidence intervals use a class-preserving crossed paired bootstrap. Before model training, 10,000 label-only replicates are generated from the locked test manifests. Each patient is assigned a stratum defined by its complete split-by-class contribution vector across all test appearances; this vector can contain several classes for patch targets. Within every nonempty stratum, the observed number of patients is resampled with replacement. A selected patient's multiplicity is then applied to all of that patient's slides and patches in every split, so overlapping test appearances never become independent people. When patient and slide coincide, slides are the resampling units. Preserving every observed contribution stratum guarantees that each class represented in an original split remains represented in every replicate; an uncomputable split-level metric is treated as an implementation failure and the analysis stops rather than silently discarding the replicate.

The label-only preflight verifies class representation, computability of every split-level metric, and identical patient multiplicities across split appearances. It also checks whether a nominally large bootstrap replicate is effectively dominated by only a few patients. If patient i has class-specific bootstrap weight v_i, its *Kish effective patient count* is

$$n_{\mathrm{Kish}} = \frac{\left(\sum_i v_i\right)^2}{\sum_i v_i^2}. \tag{14}$$

This equals the number of resampled patients when their weights are equal and becomes smaller as the weight concentrates on a few patients. The preflight records this count, the number of unique resampled patients, and the largest normalized patient weight by split and class. If $n_{\mathrm{Kish}} < 5$ in any replicate, or one patient supplies more than 50% of a class's bootstrap weight in more than 5% of replicates, inference for that dataset–regime is designated descriptive before training. The diagnostic report and resampling seed are frozen with the analysis manifest.

In the model-based bootstrap, the three split seeds are fixed design repetitions and are averaged with equal weight, not resampled as independent cohorts. Tail assignments remain separate unless an explicitly assignment-averaged estimand is reported. Within each replicate, the five matched confirmation initialization indices are resampled as paired algorithmic-noise blocks and averaged before split aggregation; they never increase the patient sample size. Balanced CE, imbalanced CE, and each method use the same patient and initialization draws. The shared balanced baseline is loaded once and receives the same bootstrap weights in every assignment-specific deficit. The full deficit and recovery ratio in Equation (13), including its numerator and denominator, is recomputed inside every replicate. Patches are never bootstrapped independently. Percentile 95% intervals are reported with the number of replicates and seed fixed in the analysis manifest.

The tested effect is gate-specific. On the discrimination axis it is the raw balanced-accuracy difference $\Delta_{\mathrm{BA}} = \mathrm{BA}(\text{method}) - \mathrm{BA}(\text{imbalanced CE})$; on the calibration axis it is the paired tail-group NLL change $\Delta_{\mathrm{cal}} = \mathrm{NLL}_{\mathrm{tail}}(\text{imbalanced CE}) - \mathrm{NLL}_{\mathrm{tail}}(\text{method})$, oriented so positive values indicate improvement. Recovery ratios are reported as normalized effects and are never test statistics. Raw two-sided p-values use a paired patient-block permutation test. Under each permutation, the complete method and CE prediction blocks are swapped within a patient using the same swap for all slides, patches, split appearances, and matched confirmation seeds, after which the gate's statistic is recomputed. All permutations are enumerated when feasible; otherwise 100,000 random permutations and the plus-one correction are used.

To keep the confirmatory family commensurate with the available independent-patient evidence, a prespecified *primary* method set is confirmatory and the remaining roster is exploratory. The primary set contains the loss-reweighting, prior-correction, and resampling references that directly target tail support and are shared across both regimes: weighted CE, focal loss, train-time logit adjustment, and balanced sampling. The confirmatory family for Holm adjustment is every primary-method $\times$ gate $\times$ severity $\times$ tail-assignment comparison within a dataset and regime; severities are not separate families, and each co-primary gate contributes its own statistic. Holm's procedure adjusts the permutation p-values across that family, with gated-out comparisons recorded as not tested. All other methods (hybrid objectives, CFAL, OKO, cRT, RankMix-inspired, SC-MIL, MDE-inspired, and post-hoc logit adjustment) are reported with the same effect estimates and intervals but form a separate exploratory family outside the confirmatory correction. Effect estimates and intervals remain primary throughout; adjusted p-values are supporting evidence. Secondary classwise, calibration, and cost analyses are exploratory unless explicitly prespecified above.

3.3 Predictors of damage and mitigation headroom

Two distinct frozen-feature measurements are made before mitigation fitting, and no predictor is derived from test examples or test outcomes. *Intrinsic separability* asks whether the classes are distinguishable in the shared Virchow2 representation before a benchmark classifier is trained. It is measured once per dataset, regime, and split using a fixed balanced reference subset drawn from the eligible training pool and shared by every condition. *Condition-specific learnability* instead asks how informative the representation remains under the exact support available in one controlled training manifest. It uses that manifest as its neighbour reference or probe-fitting set.

Both concepts are summarized on the unchanged validation set, which retains its natural class distribution. Balanced k-nearest-neighbour (kNN) macro recall predicts from the labels of the k closest reference embeddings and gives every class equal weight. A *linear probe* is a class-balanced linear classifier fitted on frozen embeddings; it measures information accessible without a flexible nonlinear head. Per-class nearest-neighbour error exposes which classes overlap locally. These diagnostics never use test labels or CE test performance. Patch probes operate on patch embeddings. WSI probes use the mean and componentwise standard deviation of all eligible instances, fixed non-learned summaries that do not inherit the attention-MIL model being explained.

Many patches from one patient are not equivalent to the same number of independent patients. The patch sensitivity analysis therefore estimates a one-way random-effects intraclass correlation (ICC) of the cross-fitted class margin. For class c, let G_c be the number of patient or slide clusters, m_{gc} their sizes, and $N_c = \sum_g m_{gc}$. With between-cluster and within-cluster mean squares $\mathrm{MS}_{B,c}$ and $\mathrm{MS}_{W,c}$, the unequal-cluster-size correction and ICC are

$$m_{0,c} = \frac{N_c - \sum_{g=1}^{G_c} m_{gc}^2/N_c}{G_c - 1}, \qquad \mathrm{ICC}_c = \frac{\mathrm{MS}_{B,c} - \mathrm{MS}_{W,c}}{\mathrm{MS}_{B,c} + (m_{0,c} - 1)\mathrm{MS}_{W,c}}. \tag{15}$$

The estimate is clipped to $[0, 1]$: zero means that patches within a cluster are no more alike than patches from different clusters, whereas one indicates complete within-cluster redundancy. The corresponding effective sample size is

$$N_{\mathrm{eff},c} = \frac{N_c}{1 + (\overline{m}_c - 1)\,\mathrm{ICC}_c}, \tag{16}$$

where $\overline{m}_c$ is the arithmetic mean cluster size. The scalar margin is the fixed intrinsic linear probe's logit for class c minus its largest competing logit. It is evaluated by *cross-fitting*: each observation is scored only by a probe trained without it. Unique patient and slide counts remain the primary support measures. Equations (15)–(16) provide a secondary sensitivity measure for patch targets, not a replacement for independent-unit counts. Slide count itself is used for WSI targets, with patient count also entered when patients contribute multiple slides.

RQ3 uses three linked exploratory analyses. Let $G_j = 1$ when comparison unit j passes either gate, let $\widehat{D}_j$ be its estimated balanced-accuracy deficit with bootstrap standard error $s_{D,j}$, and let R_j be recovery for a gated unit. With standardized log imbalance $x_{\rho,j}$, standardized intrinsic separability $x_{S,j}$, and dataset–target random intercept $b_{g[j]}$, the models have the common structure

$$\begin{aligned} \operatorname{logit} \Pr(G_j = 1) &= \alpha_G + b_{G,g[j]} + \beta_{G,\rho} x_{\rho,j} + \beta_{G,S} x_{S,j}, \\ \widehat{D}_j &\sim \mathcal{N}\big(\alpha_D + b_{D,g[j]} + \beta_{D,\rho} x_{\rho,j} + \beta_{D,S} x_{S,j},\ s_{D,j}^2 + \sigma_D^2\big), \\ R_j \mid G_j = 1 &\sim \mathcal{N}\big(\alpha_R + b_{R,g[j]} + \beta_{R,\rho} x_{\rho,j} + \beta_{R,S} x_{S,j},\ \sigma_R^2\big). \end{aligned} \tag{17}$$

The first line models whether measurable damage occurs, the second uses every signed deficit—including negative values—and the third describes recovery only after damage has been demonstrated. Adding $s_{D,j}^2$ to the residual variance prevents a noisy estimated deficit from being treated as an exact observation. This sequence is used instead of forcing occurrence, magnitude, and recovery into one hurdle model.

Because the independent unit for RQ3 is the dataset–target group and only a few such groups exist, the analyses are deliberately restricted rather than predictive. Each model uses only the two predictors shown in Equation (17). Condition-specific learnability, log minimum independent support (patients for patch targets and slides for MIL targets), and the patch-versus-WSI indicator are reported descriptively and entered only in prespecified one-predictor sensitivity fits. They are not combined because, with so few groups, they are strongly confounded with the primary predictors and with one another. Random intercepts allow each dataset–target group to have a different baseline without estimating a separate unrestricted fixed effect. The recovery model deliberately omits per-method effects, which the available gated units cannot identify.

Maximum a posteriori (MAP) estimation combines model fit with weak zero-centered, unit-scale Gaussian priors on standardized slopes and on the log standard deviations of the random intercept and residual terms. These priors regularize extreme estimates in the small sample; no interactions are fitted. Leave-one-dataset–target-group-out cross-validation repeatedly withholds one complete group and is reported only as a coarse stability check. With this small group count, held-out predictions shrink toward the population-average intercept, so RQ3 remains a hypothesis-generating description of association rather than a validated predictive model. TCGA-UT supplies the only genuine paired-regime evidence; the CAMELYON16, PANDA, and BRACS targets differ by regime. Model definitions, priors, and covariate roles are frozen before recovery outcomes are inspected, and all RQ3 results remain exploratory.

A Experimental Controls

Every trainable method uses the common learning-rate grid

$$\eta \in \{1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 3 \times 10^{-3}\}. \tag{18}$$

For methods in Table 4, this grid is crossed with the four candidate values of the method-specific control. The resulting maximum is 16 configurations per method, dataset, regime, and severity. Any computationally necessary narrowing is completed and recorded in the signed analysis manifest before the first definitive fit; no grid is changed once definitive training begins.

Method or family	Candidate method-specific control
Balanced sampling / weighted CE	strength $\in \{0.25, 0.5, 0.75, 1\}$
Focal loss	$\gamma \in \{0.5, 1, 1.5, 2\}$
Logit adjustment	$\tau \in \{0.25, 0.5, 1, 2\}$
CE + soft F1 / soft MCC	auxiliary weight $\in \{0.25, 1, 4, 16\}$
CFAL	affinity bandwidth $\sigma \in \{0.25, 1, 2, 4\}$
OKO	odd-class count $k \in \{1, 2, 4, 8\}$, capped by $K - 1$
RankMix-inspired	beta shape $\alpha \in \{0.5, 1, 2, 4\}$
SC-MIL	temperature $\tau \in \{0.05, 0.1, 0.2, 0.5\}$
MDE-inspired	consistency weight $\lambda_{\text{con}} \in \{0, 0.1, 0.25, 0.5\}$

Table 4: Prespecified method-specific grids crossed with the common learning-rate grid in Equation (18). cRT has no imbalance-strength grid; only its stage-two learning rate is selected separately.

For every realized condition, the experiment record contains dataset version; eligibility rules; patient, slide, and patch identifiers by partition; requested and achieved T and ρ; class assignment; all seed families and their tuning, confirmation, pilot, or definitive roles; pilot patch quota; bootstrap-preflight diagnostics; model and optimizer configuration; candidate grid; selected checkpoint; environment identifier; and test-prediction hash. Deviations are dated and justified before the affected test labels are accessed.

B Terminology

Natural anchor. The full eligible natural training partition. It is descriptive and may have a different size from controlled conditions.

Size-matched balanced reference. A training subset with total support T and approximately equal class counts; it defines the counterfactual reference in D_M.

Imbalance severity. The requested training allocation summarized by ρ; achieved severity is computed from realized integer counts.

Pilot level. A candidate number of independent training units per class used in the balanced CE support-stability study.

Tail assignment. The locked mapping from semantic classes to positions in the constructed support profile.

Comparison unit. One dataset, prediction regime, imbalance severity, and tail assignment evaluated together.

Deficit gate. A prespecified threshold rule that determines whether an observed discrimination or calibration deficit is large and precise enough for recovery analysis.

Imbalance deficit. The paired balanced-CE minus imbalanced-CE difference at common T and on the same held-out cases.

Recovery. A method's improvement over imbalanced CE divided by the evidenced imbalance deficit.

Kish effective patient count. The equally weighted patient count with the same weight concentration as one bootstrap replicate.

Intraclass correlation. The estimated similarity of class margins from patches belonging to the same patient or slide cluster.

Patch-micro estimate. A metric computed over eligible patch predictions, with patient-clustered uncertainty.

Patient-macro estimate. An equal-weight aggregation over slides or patients, limiting domination by heavily tiled cases.

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